

**Department of Biomedical Engineering Technology
The Superior University, Lahore**

Computer Programming

**Experiment No.5
Conditional Statements and Loops**

Prepared for:

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Section: Biomedical Engineering__

Semester: Second_____

Total Marks: _____10_____

Obtained Marks: _____

Signature: _____

Date: _____

Experiment No.5

Conditional Statements and Loops

(Rubrics)

Name: Mazen Abdulgani_____

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A. PSYCHOMOTOR

Serial No.	Criteria	Allocated Marks	Unacceptable 0%	Poor 25%	Fair 50%	Good 75%	Excellent 100%	Total Obtained
1	Syntax Handling	1	0	0.25	0.5	0.75	1	
2	Program Coding Skills	2	0	0.5	1	1.5	2	
3	Accuracy in Output Results	3	0	0.75	1.5	2.25	3	
Sub-Total		6	Sub Total Marks Obtained in Psychomotor (P)					

B. AFFECTIVE

Serial No.	Criteria	Allocated Marks	Unacceptable 0%	Poor 25%	Fair 50%	Good 75%	Excellent 100%	Total Obtained
1	Assigned tasks	2	0	0.5	1	1.5	2	
2	Lab Report	2	0	0.5	1	1.5	2	
Sub-Total		4	Sub Total Marks Obtained in Affective (A)					

Instructor Name: Miss Eisha NawazTotal Marks (P+A): 10

Instructor Signature: _____

Obtained Marks (P+A): _____

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Ladder If, Switch Statement and While Loop

1 OBJECTIVES

- To get basic understanding of logical operator.
- To practice how to use the if-else ladder.
- To get basic understanding of switch statement.
- Where and when to use 'Switch Statement'.
- To understand the basics of WHILE Loop.

2 LOGICAL OPERATORS

The logical operators `&&` and `||` are used when evaluating two expressions to obtain a single relational result. The operator `&&` corresponds to the Boolean logical operation AND, which yields true if both its operands are true, and false otherwise. The following panel shows the result of operator `&&` evaluating the expression `a&&b`:

&& Operator (AND)		
a	b	a&&b
True	False	False
True	True	True
False	False	False
False	True	False

The operator `||` corresponds to the Boolean logical operation OR, which yields true if either of its operands is true, thus being false only when both operands are false. Here are the possible results of `a||b`:

 Operator (OR)		
a	b	a b
True	False	True
True	True	True
False	False	False
False	True	True

2.1 SYNTAX

```
((5 == 5) && (3 > 6)) // evaluates to false (true && false)
```

```
( (5 == 5) || (3 > 6) ) // evaluates to true ( true || false )
```

When using the logical operators, C++ only evaluates what is necessary from left to right to come up with the combined relational result, ignoring the rest. Therefore, in the last example ((5==5) || (3>6)), C++ evaluates first whether 5==5 is true, and if so, it never checks whether 3>6 is true or not. This is known as short-circuit evaluation, and works like this for these operators:

2.2 EXAMPLE 1

Code	Output
<pre>//Program to check if a number is greatest among three numbers using logical && operator int main () { int a,b,c; cout<<" enter three numbers\n"; cin>>a>>b>>c; if(a>b && a>c) cout<<"A is greatest.\n"; else cout<<"A is not greatest\n"; return 0; }</pre>	<pre>Enter three number: 15 12 11 A is greatest</pre>

2.3 EXAMPLE 2

Code	Output
<pre>#include <iostream> using namespace std; int main() { int a = 5, b = 2, c = 6, d = 4; if (a == b && c > d) cout << "a equals to b AND c is greater than d\n"; else cout << "AND operation returned false\n"; if (a == b c > d) cout << "a equals to b OR c is greater than d\n"; else cout << "Neither a is equal to b nor c is greater than d\n"; if (!b) cout << "b is zero\n"; else</pre>	<pre>AND operation returned false A equals to b OR c is greater than d B is not zero</pre>

<pre> cout << "b is not zero"; return 0; } </pre>	
--	--

3 IF...ELSE LADDER

The if...else statement executes two different codes depending upon whether the test expression is true or false. Sometimes, a choice has to be made from more than 2 possibilities.

The if...else ladder allows you to check between multiple test expressions and execute different statements.

3.1 SYNTAX

```

if (test expression1) {
// statement(s)
}
else if (test expression2) {
// statement(s)
}
else if (test expression3) {
// statement(s)
}
.
.
else {
// statement(s)
}

```

3.2 EXAMPLE 1

Code	Output
<pre> #include <iostream> using namespace std; int main () { int number1, number2; cout<<"Enter two integers: "<<endl; cin>>number1>>number2; //checks if the two integers are equal. if (number1 == number2) { cout<<"Result:"<<number1<<"=="<<number2<<endl; } //checks if number1 is greater than number2. else if (number1 > number2) { cout<<"Result:"<<number1<<">"<<number2<<endl; </pre>	<pre> Enter two integers: 12 23 Result: 12 < 23 </pre>

```
}  
//checks if both test expressions are false  
else {  
cout<<"Result:"<<number1<<" "<<number2<<endl;  
}  
return 0;  
}
```

4 SWITCH STRUCTURE:

Recall that there are two selections, or branch, structures in C++. The first selection structure, which is implemented with if and if. .else statements, usually requires the evaluation of a (logical) expression. The second selection structure, which does not require the evaluation of a logical expression, is called the switch structure. C++'s switch structure gives the computer the power to choose from among many alternatives.

A general syntax of the switch statement is:

```
switch (expression)  
{  
case value1:  
statements1  
break;  
case value2:  
statements2  
break;  
.  
.  
case value n:  
statements n  
break;  
default:  
statements  
}
```

In C++, switch, case, break, and default are reserved words. In a switch structure, first the expression is evaluated. The value of the expression is then used to perform the actions specified in the statements that follow the reserved word case. Recall that in a syntax, shading indicates an optional part of the definition.

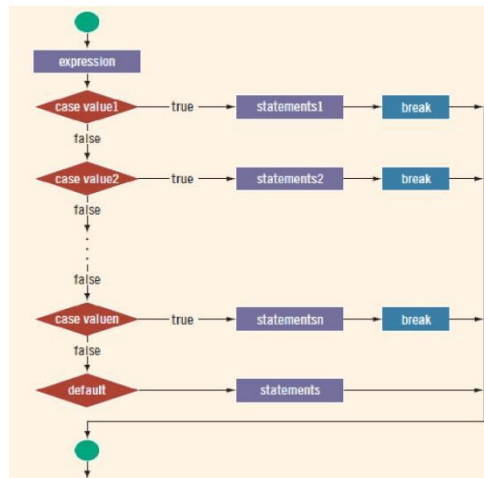


Figure 1 Switch Statement

In example 1, we use CHAR TYPE variable. It means in the CASE CRITERIA we will write 'A', 'B', etc. using COMMAS ' ' just like we declare a CHAR Type variable (For example char word= 'C')

4.1 EXAMPLE 1

Code	Output
<pre> //Switch Example 1 #include<iostream> using namespace std; int main () { char grade; cout<<"Enter your grade: "; cin>>grade; switch (grade) { case 'A': 3 cout<< "The grade point is 4.0."; break; case 'B': cout<< "The grade point is 3.0."; break; case 'C': cout<< "The grade point is 2.0."; break; case 'D': cout<< "The grade point is 1.0."; break; case 'F': cout<< "The grade point is 0.0."; </pre>	

<pre> break; default: cout<< "The grade is invalid."; } return 0; } </pre>	
---	--

In example 1, we use INTEGER TYPE variable. It means in the CASE CRITERIA we will write 1, 2, 3, 100, etc. without using COMMAS ‘ ‘ just like we declare an INTEGER Type variable (For example int a= 3)

4.2 EXAMPLE 2

Code	Output
<pre> //Switch Example 2 #include <iostream> using namespace std; int main () { int score; cout<<"Enter your score (Max 5 Marks) : "; cin>>score; switch (score) { case 0: cout<<"Your Score is 0"; break; case 1: cout<<"Your Score is 1"; break; case 2: cout<<"Your Score is 2"; break; case 3: cout<<"Your Score is 3"; break; case 4: cout<<"Your Score is 4"; break; case 5: cout<<"Your Score is 5"; break; default: cout<< "Invalid test score." <<endl; } } </pre>	

```
}  
}
```

5 LOOPS IN PROGRAMMING

In computer programming, loops are used to repeat a block of code.

For example, let's say we want to show a message 100 times. Then instead of writing the print statement 100 times, we can use a loop.

That was just a simple example; we can achieve much more efficiency and sophistication in our programs by making effective use of loops.

5.1 TYPES OF LOOPS

There are three types of loops in C++ as shown in the given figure:

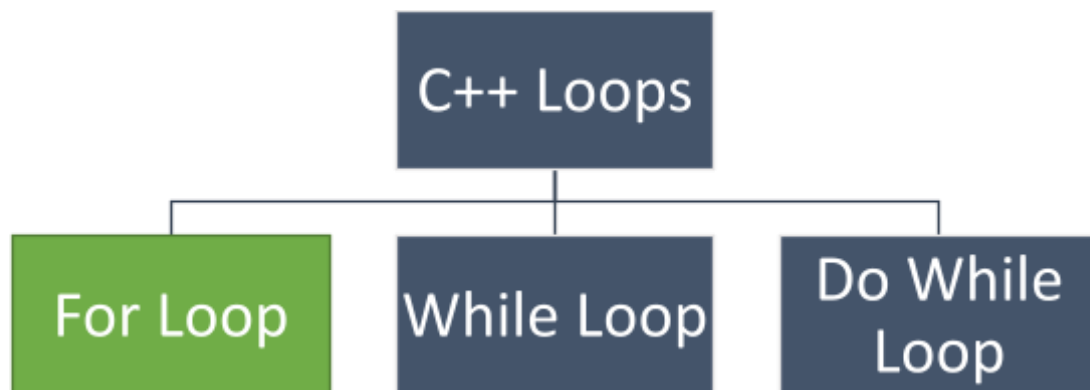


Figure 2 Types of Loops in C++

5.2 C++ WHILE LOOP

The syntax of while loop is as follows:

```
while (condition)  
{  
    // body of the loop  
}
```

Here,

1. A while loop evaluates the condition
2. If the condition evaluates to true, the code inside the while loop is executed.

3. The condition is evaluated again.
4. This process continues until the condition is false.
5. When the condition evaluates to false, the loop terminates.

5.3 FLOWCHART OF WHILE LOOP

The given figure shows the flowchart of the while loop.

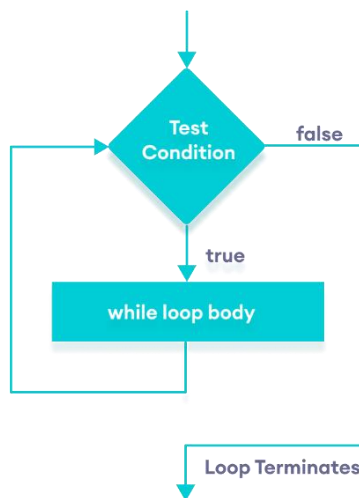


Figure 3s Flowchart of while Loop

5.4 EXAMPLE

A program to print numbers from 1 to 5

Code	Output
<pre> #include <iostream> using namespace std; int main () { int i = 1; while(i<=5){ cout << i << " "; ++i; } return 0; } </pre>	1 2 3 4 5

Here is how this program works:

Iteration	Variable	i <= 5	Action
1st	i = 1	true	1 is printed. i is increased to 2.
2nd	i = 2	true	2 is printed. i is increased to 3.
3rd	i = 3	true	3 is printed. i is increased to 4.
4th	i = 4	true	4 is printed. i is increased to 5.
5th	i = 5	true	5 is printed. i is increased to 6.
6th	i = 6	false	The loop is terminated

5.5 EXAMPLE 2

Write a program to display Sum of positive numbers only

Code	Output
<pre>// program to find the sum of positive numbers // if the user enters a negative number, the loop ends // the negative number entered is not added to the sum #include <iostream> using namespace std; int main() { int number; int sum = 0; // take input from the user cout << "Enter a number: "; cin >> number; while (number >= 0) { // add all positive numbers sum += number; // take input again if the number is positive cout << "Enter a number: "; cin >> number; } // display the sum cout << "\nThe sum is " << sum << endl; return 0; }</pre>	<pre>Enter a number: 6 Enter a number: 12 Enter a number: 7 Enter a number: 0 Enter a number: -2 The sum is 25</pre>

In this program, the user is prompted to enter a number, which is stored in the variable number.

- In order to store the sum of the numbers, we declare a variable sum and initialize it to the value of 0.
- The while loop continues until the user enters a negative number. During each iteration, the number entered by the user is added to the sum variable.
- When the user enters a negative number, the loop terminates. Finally, the total sum is displayed.

5.6 EXAMPLE 3

Program to print sum of first N natural numbers using while loop in C++:

Code	Output
<pre>#include <iostream> using namespace std; int main() { int n, i = 1, sum = 0; cout << "Enter number:" << endl; cin >> n; cout << endl; while (i <= n) { sum += i; i++; } cout << "Sum N no. is " << sum << endl; return 0; }</pre>	

6 LAB TASKS

6.1 TASK 1

Patient Monitoring System: Simulate a basic hospital monitoring unit where different vital signs are selected and continuously evaluated to classify a patient's condition.

Requirements

1. Use a **while loop** to continuously run the system until user exits
2. Use a **switch** statement to select:
 - 1 → Heart Rate

Heart Rate (bpm)	Condition	Output Message
< 60	Low	Bradycardia
60 – 100	Normal	Normal Heart Rate
> 100	High	Tachycardia

- 2 → Temperature

Temperature (°C)	Condition
< 36	Low (Hypothermia)
36 – 37.5	Normal
> 37.5	High (Fever)

- 3 → Exit
3. Use **if-else-if** to classify:
 - Heart Rate → Low (Bradycardia) / Normal / High (Tachycardia)
 - Temperature → Low (Hypothermia) / Normal / High (Fever)

Logic

1. Menu:
 - 1 → Heart Rate
 - 2 → Temperature

- 3 → Exit
2. Loop until exit
 3. Apply condition-based classification

Starter Code (C++)**Starter Code (Complete the TODO sections)**

```
//Student Name
//Student Roll Number

#include <iostream>
using namespace std;

int main() {
    //TODO starter cout section

    // TODO your code here

    return 0;
}
```

Expected Output**Expected Output**

```
=====
PATIENT MONITORING SYSTEM
=====

1. Heart Rate
```

```
2. Temperature
3. Exit
Enter choice: 1

Enter heart rate: 110
Result:
Tachycardia detected (High)
```

```
1. Heart Rate
2. Temperature
3. Exit
Enter choice: 2

Enter body temperature (°C): 36.5
Result:
Normal Temperature
```

```
1. Heart Rate
2. Temperature
3. Exit
Enter choice: 3
System Exit
```



```

1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5      int choice;
6      bool running = true;
7
8      while (running)
9      {
10         cout<<"-----PATIENT MONITORING SYSTEM-----\n";
11         cout<<"1- Heart Rate\n";
12         cout<<"2- Temperature\n";
13         cout<<"3- Exit\n";
14         cout<<"Enter your choice: ";
15         cin>> choice;
16
17         switch (choice)
18         {
19             case 1:
20                 int heartrate;
21                 cout<<"Enter Heart Rate (bpm): ";
22                 cin>> heartrate;
23
24                 if (heartrate < 60){
25                     cout<<"Condition: Low\n";
26                     cout<<"Output Message: Bradycardia\n\n";
27                 }
28                 else if (heartrate >= 60 && heartrate <= 100){
29                     cout<<"Condition: Normal\n";
30                     cout<<"Output Message: Normal Heart Rate\n\n";
31                 }
32                 else {
33                     cout<<"Condition: High\n";
34                     cout<<"Output Message: Tachycardia\n\n";
35                 }
36                 break;
37             case 2:
38                 float temperature;
39                 cout<<"Enter Temperature (°C): ";
40                 cin>> temperature;
41
42                 if (temperature < 36) {
43                     cout << "Condition: Low (Hypothermia)\n\n";
44                 }
45                 else if (temperature >= 36 && temperature <= 37.5) {
46                     cout << "Condition: Normal\n\n";
47                 }
48                 else {
49                     cout << "Condition: High (Fever)\n\n";
50                 }
51                 break;
52             case 3:
53                 cout << "Exiting Patient Monitoring System...\n\n";
54                 running = false;
55                 break;
56             default:
57                 cout << "Invalid choice! Please try again.\n";
58         }
59     }

```

```

-----PATIENT MONITORING SYSTEM-----
1- Heart Rate
2- Temperature
3- Exit
Enter your choice: 1
Enter Heart Rate (bpm): 103
Condition: High
Output Message: Tachycardia

-----PATIENT MONITORING SYSTEM-----
1- Heart Rate
2- Temperature
3- Exit
Enter your choice: 2
Enter Temperature (°C): 35.6
Condition: Low (Hypothermia)

-----PATIENT MONITORING SYSTEM-----
1- Heart Rate
2- Temperature
3- Exit
Enter your choice: 3
Exiting Patient Monitoring System...

```

7 HOME ASSIGNMENT:

7.1 TASK 1

Smart ICU Alert System: Develop a smart ICU system that continuously monitors patient type and vital values to generate alerts based on risk levels.

Students must take one physiological input: **Oxygen Saturation** (SpO₂ %)

Requirements:

1. while loop → continuous monitoring
 - Program must keep asking for SpO₂ values
 - Stop only when user enters -1
2. switch → patient type:
 - 1 → Adult
 - 2 → Child

Patient Type	Code	Description
Adult	1	Age ≥ 18
Child	2	Age < 18

3. if-else-if → risk detection (Normal / Warning / Critical)

For Adult Patient:

SpO ₂ Range	Condition
≥ 95	Normal
90 – 94	Warning
< 90	Critical

For Child Patient:

SpO ₂ Range	Condition
≥ 96	Normal
92 – 95	Warning
< 92	Critical

4. Alert Logic

Condition	Output Message
Normal	“Patient Stable”
Warning	“Warning: Monitor Closely”
Critical	“CRITICAL ALERT! Immediate Action Required”

Logic

1. Display header
2. Enter loop → while (value != -1)
3. Ask patient type → switch
4. Inside case:
 - Input SpO₂
 - Apply if–else-if thresholds
 - Print condition
5. Repeat

Starter Code (C++)**Starter Code (Complete the TODO sections)**

```
//Student Name
//Student Roll Number

#include <iostream>
using namespace std;

int main() {
    //TODO starter cout section

    // TODO Write your code here

    return 0;
}
```

Expected Output

Expected Output

```
=====
SMART ICU ALERT SYSTEM
```

```
=====
```

```
Select Patient Type:
```

```
1. Adult
```

```
2. Child
```

```
Enter choice (-1 to exit): 1
```

```
Enter SpO2 level: 93
```

```
Result:
```

```
Warning: Monitor Closely
```

```
-----  
Select Patient Type:
```

```
1. Adult
```

```
2. Child
```

```
Enter choice (-1 to exit): 2
```

```
Enter SpO2 level: 85
```

```
Result:
```

```
CRITICAL ALERT! Immediate Action Required
```

```
-----  
Select Patient Type:
```

```
1. Adult
```

```
2. Child
```

```
Enter choice (-1 to exit): -1
```

```
System Exit
```

```

1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5      int choice;
6      int SpO2;
7      bool running = true;
8
9      while (running)
10     {
11         cout<<"-----SMART ICU ALERT SYSTEM-----\n";
12         cout<<"Select Patient Type: \n";
13         cout<<"1- Adult\n";
14         cout<<"2- Child\n";
15         cout<<"Enter choice: \n";
16         cin>> choice;
17
18         switch (choice)
19         {
20             case 1:
21                 float SpO2;
22                 cout<<"Enter SpO2 level: \n";
23                 cin>>SpO2;
24
25                 if (SpO2 >= 96){
26                     cout<<"Condition: Normal";
27                     cout<<"Output Message:Patient Stable\n";
28                 }
29                 else if (SpO2 >= 90 && SpO2 <= 94)
30                 {
31                     cout<<"Condition: Warning";
32                     cout<<"Output Message: Warning: Monitor Closely\n";
33                 }
34                 else
35                 {
36                     cout<<"Critical";
37                     cout<<"CRITICAL ALERT!! Immediate Action Required\n";
38                 }
39                 break;
40             case 2:
41                 cout<<"Enter SpO2: \n";
42                 cin>> SpO2;
43
44                 if (SpO2 >= 96)
45                 {
46                     cout<<"Condition: Normal";
47                     cout<<"Output Message:Patient Stable\n";
48                 }
49                 else if (SpO2 >= 92 && SpO2 <= 95)
50                 {
51                     cout<<"Condition: Warning";
52                     cout<<"Output Message: Warning: Monitor Closely\n";
53                 }
54                 else
55                 {
56                     cout<<"Critical";
57                     cout<<"CRITICAL ALERT!! Immediate Action Required\n";
58                 }
59                 break;
60         }
61     }
62 }

```

===== Smart ICU Alert System =====

Select Patient Type:

1. Adult

2. Child

Enter choice: 1

Enter SpO2 value (-1 to stop): 97

Patient Type: Adult

Condition: Normal

===== Smart ICU Alert System =====

Select Patient Type:

1. Adult

2. Child

Enter choice: 2

Enter SpO2 value (-1 to stop): 90

Patient Type: Child

Condition: Critical